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#include <stdio.h>

#define People 100

int array[People];

int findnextk(int s, int k, int n)//this accept the k and n from the main function, but also the P.
{
    while (k) //As long a k has a value, this loop is true, and we keep running the loop. This loop
    skipping all the lucky ones that k passes over
    {
        if (array[s=(s+1)%n]!=-1)// Now, the s=(s+1)%n part is called 'incrimenting the circle', aka, we
        are checking the array address to see if each one continue a value, or has nothing inside it at all. If
        they don't have a value inside of them, skip them.
        {
            k--; //Yay, someone' been found. decrement from 2 to 1, or what ever you set you k to.
        }
    }
    return s; //retuning s are you P position in your pain function.
}

void swap(int *x, int *y) //this functions takes in array[index] and array[P] and swaps them.
{
    int temp= *x;
    *x = *y;
    *y = temp;
}

int main()
{
    int n ,k;

    while (scanf("%d %d", &n, &k) != EOF && (n || k))
    { //Start of the while loop. This will run as long as n and k are NOT = 0.
        for (int i=0; i<n; i++)
        { //Start of for loop.
            array[i]=i; //Each position will get a number starting from 1. 0 hase 1st, 1 has 2nd, 2 has
            3rd.
        } //End of for loop.

        int P=(k-1)%n; //P=position of the person currently holding the big fuck off sword, well, at
        least the INITIAL person with the big fuck off sword. Now, I'll be honest, I'm only % by n because
        someone told me it kept the P from going outside the bounds of the array.

        for(int i=1; i<n; i++)
        { //Start of for loop
            array[P]=-1;
            int index=findnextk(P,k,n);
            swap(&array[index], &array[P]); //ya, you see that guy who died, we need to remove him,
            and have the guy next to him take his place.
            P=findnextk(P,k,n);
        } //End of for loop.

        printf("%d\n", (n-array[P])%n+1);
    }
    return 0;
}

```